Networking Guide

From Hubs & Switches to VLANs and ACLs in Cisco Packet Tracer

Networking Fundamentals throughout Practical Labs

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Part I

Foundations — Devices, LANs & Subnetting

Chapter 1

Introduction to Networking Devices

1.1 Why Networking Devices Matter

Every network — from a two-PC home setup to an enterprise campus — is built from three fundamental "connecting" devices: the **Hub**, the **Switch**, and the **Router**. Understanding exactly what each device does at which OSI layer is the single most important foundation for everything that follows (VLANs, routing, ACLs).

1.2 The Hub (Layer 1 Device)

A hub is a dumb, multi-port repeater. It operates purely at **Layer 1 (Physical)**.

- Receives an electrical signal on one port and **blindly repeats it out of every other port**.
- Has **no MAC address table** — it does not understand frames or addresses.
- All ports share **one single collision domain** and **one broadcast domain**.
- Operates in **half-duplex** only → causes collisions as more devices are added.

WARNING

Hubs are obsolete in real networks and in Packet Tracer are mainly used to *demonstrate* collision domains. Never use a hub in a design task unless specifically asked to illustrate collisions.

1.3 The Switch (Layer 2 Device)

A switch is an intelligent, multi-port bridge operating at **Layer 2 (Data Link)**.

- Learns source **MAC addresses** of incoming frames and builds a **MAC Address Table (CAM table)**.
- Forwards frames *only* to the port where the destination MAC resides (unicast) — unknown/broadcast/multicast frames are flooded.
- **Each switch port is its own collision domain** (because of full-duplex, point-to-point links).
- **All ports on a switch remain in one broadcast domain** — unless VLANs are configured (Part II).

TIP

In Packet Tracer, click a switch while frames are moving (Simulation Mode) and open its MAC table under `CLI > show mac address-table` to visually understand learning.

1.4 The Router (Layer 3 Device)

A router works at **Layer 3 (Network)**, making forwarding decisions based on **IP addresses**.

- Each router interface belongs to a **different network/subnet** and is a **separate broadcast domain**.
- Maintains a **Routing Table** (built via connected routes, static routes, or dynamic routing protocols).
- Used to connect multiple LANs/VLANs together, and connect LANs to the Internet/WAN.

Device	OSI Layer	Collision Domain	Broadcast Domain
Hub	1 (Physical)	1 per hub	1 per hub
Switch	2 (Data Link)	1 per port	1 per switch (no VLAN)
Router	3 (Network)	1 per port	1 per port/interface

Chapter 2

Building LAN in Packet Tracer

2.1 What Is a LAN?

A **Local Area Network (LAN)** is a group of devices (PCs, printers, servers) connected together, typically within one building/location, sharing the **same subnet/broadcast domain**, usually via a switch.

2.2 LAN Using a Switch Only

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Topology: 3 PCs → connected to **Switch2960** using straight-through (copper) cables.
Steps in Packet Tracer:

1. Drag 1 Switch (2960) and 3 PCs onto the workspace.
2. Connect PC0, PC1, PC2 to switch FastEthernet0/1, 0/2, 0/3 using **Automatic** or **Copper Straight-through** cable.
3. Click each PC → Desktop → IP Configuration → assign:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

TIP

Test connectivity using Command Prompt on any PC:

```
C:\> ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

2.3 LAN Using a Router (Connecting Two Separate LANs)

A single switch LAN cannot talk to another LAN on a different subnet without a router.

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Topology:

- LAN1: PC0, PC1 → Switch0 → Router G0/0/0
- LAN2: PC2, PC3 → Switch1 → Router G0/0/1

CHEAT SHEET: Basic Router Interface Configuration

```
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# exit
Router# write memory      ! saves running-config to startup-config
```

NOTE

Each PC must use the router's connected interface IP as its **Default Gateway**:

- PC0, PC1 Gateway = 192.168.1.1
- PC2, PC3 Gateway = 192.168.2.1

Without this gateway, inter-LAN communication will fail even if IPs are correctly set.

2.4 Verifying the Router

```
Router# show ip interface brief
Router# show running-config
Router# show ip route
```

Chapter 3

Subnetting & Subnet Masking

3.1 IP Addressing Recap

An IPv4 address is 32 bits, written as 4 octets (e.g., `192.168.1.10`). It is divided into:

- **Network portion** — identifies the LAN/subnet.
- **Host portion** — identifies the device within that subnet.

3.2 The Subnet Mask

The subnet mask tells a device which bits are "network" and which are "host" bits.

CIDR	Subnet Mask	Hosts Usable
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6
/30	255.255.255.252	2 (point-to-point link)

NOTE

Formula for usable hosts: $2^h - 2$, where h = number of host bits (the -2 removes the Network ID and Broadcast ID).

3.3 CIDR Notation

CIDR (**Classless Inter-Domain Routing**) notation expresses the mask as a slash + number of network bits, e.g. `192.168.1.0/24` = 24 network bits, 8 host bits.

3.4 Step-by-Step Subnetting Example

Problem: Subnet `192.168.10.0/24` into 4 equal subnets.

1. Number of subnets needed = 4 ⇒ borrow bits so $2^n \geq 4 \Rightarrow n = 2$ bits borrowed.
2. New mask = /24 + 2 = /**26** = 255.255.255.192

3. Block size = 256 − 192 = 64

Subnet	Network ID	Usable Range	Broadcast
1	192.168.10.0	.1 – .62	192.168.10.63
2	192.168.10.64	.65 – .126	192.168.10.127
3	192.168.10.128	.129 – .190	192.168.10.191
4	192.168.10.192	.193 – .254	192.168.10.255

TIP

Shortcut trick ("Magic Number" method): Block size = 256 − interesting octet value of mask. Keep adding the block size to get each subnet's Network ID.

3.5 VLSM (Variable Length Subnet Masking)

VLSM allows subnets of **different sizes** from the same base network — essential for efficient router/VLAN designs where one LAN needs 50 hosts and another only needs 2 (e.g., a router-to-router link).

CHEAT SHEET: Subnetting Quick Reference

```
/24 = 255.255.255.0     -> 256 addresses / 254 hosts
/25 = 255.255.255.128   -> 128 addresses / 126 hosts
/26 = 255.255.255.192   -> 64 addresses  / 62 hosts
/27 = 255.255.255.224   -> 32 addresses  / 30 hosts
/28 = 255.255.255.240   -> 16 addresses  / 14 hosts
/29 = 255.255.255.248   -> 8  addresses  / 6 hosts
/30 = 255.255.255.252   -> 4  addresses  / 2 hosts  (router links)
/32 = 255.255.255.255   -> 1 address (host route)
```

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In Packet Tracer, build the 4-subnet design above:

1. Create 4 separate switch LANs with 2 PCs each.
2. Assign IPs per the subnet table above.
3. Connect each LAN to a dedicated router interface with the matching **Network ID's first usable IP** as gateway.
4. Verify using `ping` across LANs after setting gateways correctly.

Part II

VLANs & Access Control Lists

Chapter 4

Understanding VLANs

4.1 What Is a VLAN?

A **Virtual LAN (VLAN)** logically segments a single physical switch into multiple separate broadcast domains — *without* needing separate physical switches.

- Each VLAN = its own broadcast domain = its own subnet.
- Devices in different VLANs **cannot** communicate without a Layer-3 device (router or Layer-3 switch).
- Benefits: security, reduced broadcast traffic, logical grouping (e.g., Sales VLAN, HR VLAN) regardless of physical location.

4.2 Access Ports vs Trunk Ports

- **Access Port:** Belongs to exactly one VLAN; connects to end devices (PCs, printers).
- **Trunk Port:** Carries traffic for **multiple VLANs** between switches or between switch and router, using **802.1Q tagging**.

Port Type	Used Between
Access	Switch ↔ End device
Trunk	Switch ↔ Switch, Switch ↔ Router

Chapter 5

Router-on-a-Stick

5.1 Topology

- 1 Router (single physical interface, e.g., G0/0)
- 1 Switch (2960)
- VLAN 10 (Sales) — PC0, PC1
- VLAN 20 (HR) — PC2, PC3
- Switch trunk port connects to Router G0/0 (the router interface is divided into **sub-interfaces**, hence "router-on-a-stick")

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Physical Cabling:

- PC0, PC1 → Switch Fa0/1, Fa0/2 (VLAN 10)
- PC2, PC3 → Switch Fa0/3, Fa0/4 (VLAN 20)
- Switch Gig0/1 → Router G0/0 (Trunk link)

5.2 Step 1: Create VLANs on the Switch

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
```

5.3 Step 2: Assign Access Ports to VLANs

```
Switch(config)# interface range fastEthernet0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

5.4 Step 3: Configure Trunk Port Toward Router

```
Switch(config)# interface gigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# exit
```

5.5 Step 4: Configure Router Sub-Interfaces

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface gigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

NOTE

`encapsulation dot1Q 10` must match the VLAN ID exactly. Each sub-interface acts as the **default gateway** for its VLAN.

5.6 Step 5: PC Configuration

PC	VLAN	IP	Gateway
PC0	10	192.168.10.2	192.168.10.1
PC1	10	192.168.10.3	192.168.10.1
PC2	20	192.168.20.2	192.168.20.1
PC3	20	192.168.20.3	192.168.20.1

5.7 Verification Commands

```
Switch# show vlan brief
Switch# show interfaces trunk
Router# show ip interface brief
Router# show running-config
```

Chapter 6

Switch Inter-VLAN Routing

6.1 Concept: Layer 3 Switch with SVIs

Without a router, a **Layer 3 (Multilayer) Switch** (e.g., Cisco 3560/3650 in Packet Tracer) can route between VLANs internally using **Switch Virtual Interfaces (SVIs)**.

NOTE

This method requires a **Multilayer Switch** — a normal Layer 2 switch (2960) **cannot** route between VLANs on its own.

6.2 Topology

- 1 Multilayer Switch (e.g., 3560)
- VLAN 10 — PC0, PC1 on Fa0/1, Fa0/2
- VLAN 20 — PC2, PC3 on Fa0/3, Fa0/4
- No external router needed

6.3 Step 1: Create VLANs & Assign Ports

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

6.4 Step 2: Enable Routing & Create SVIs

```
Switch(config)# ip routing
! Critical step - turns on Layer 3 routing on the switch

Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

WARNING

Forgetting `ip routing` is the #1 mistake — without it, SVIs will have "up/up" status but will **not route** between VLANs.

6.5 Step 3: PC Gateway Configuration

PC	VLAN	IP	Gateway
PC0	10	192.168.10.2	192.168.10.1
PC1	10	192.168.10.3	192.168.10.1
PC2	20	192.168.20.2	192.168.20.1
PC3	20	192.168.20.3	192.168.20.1

6.6 Verification

```
Switch# show ip route
Switch# show vlan brief
Switch# show ip interface brief
```

CHEAT SHEET: VLAN Commands Summary

```
vlan <id>                          ! create VLAN
name <name>                        ! name the VLAN
switchport mode access             ! set port as access
switchport access vlan <id>        ! assign VLAN to port
switchport mode trunk              ! set port as trunk
switchport trunk allowed vlan <list>
interface vlan <id>                ! create SVI (L3 switch)
ip routing                         ! enable L3 switch routing
encapsulation dot1Q <id>           ! router sub-interface tagging
show vlan brief
show interfaces trunk
show ip interface brief
```

Chapter 7

Access Control Lists (ACLs)

7.1 What Is an ACL?

An **Access Control List** is a set of rules applied to a router (or L3 switch) interface to **permit or deny** traffic based on defined criteria. Processed **top-down**, first match wins, and there is an **implicit deny all** at the end of every ACL.

7.2 Standard ACL

7.2.1 Characteristics

- Number range: **1–99** and **1300–1999** (expanded range)
- Filters based **only on Source IP Address**
- Best practice: place **close to the destination** (since it can't filter by destination/port, placing it near source would block too much).

7.2.2 Syntax

```
Router(config)# access-list <1-99> {permit|deny} <source-ip> <wildcard-mask>
```

7.2.3 Example: Block PC on 192.168.10.5 from reaching network 192.168.20.0/24

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Topology: Two LANs connected via router as in Chapter 2. Goal: deny host 192.168.10.5, allow everyone else.

```
Router> enable
Router# configure terminal
Router(config)# access-list 10 deny host 192.168.10.5
Router(config)# access-list 10 permit any
Router(config)# exit
```

7.2.4 Applying to an Interface

```
Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip access-group 10 out
Router(config-if)# exit
```

NOTE

Wildcard masks are the inverse of subnet masks. E.g., for network 192.168.20.0/24, wildcard = 0.0.0.255.
Shortcut: `host 192.168.10.5` = `192.168.10.5 0.0.0.0`.
`any` = `0.0.0.0 255.255.255.255`

7.3 Extended ACL

7.3.1 Characteristics

- Number range: **100–199** and **2000–2699** (expanded range)
- Filters based on **Source IP, Destination IP, Protocol (TCP/UDP/ICMP), and Port Numbers**
- Best practice: place **close to the source** (filters precisely, so blocking early saves bandwidth).

7.3.2 Syntax

```
Router(config)# access-list <100-199> {permit|deny} <protocol> \
   <source-ip> <wildcard> <destination-ip> <wildcard> [eq <port>]
```

7.3.3 Example: Block only HTTP (Web) from LAN1 to Server, allow everything else

```
Router(config)# access-list 110 deny tcp 192.168.10.0 0.0.0.255 \
   host 192.168.20.100 eq 80
Router(config)# access-list 110 permit ip any any
```

7.3.4 Applying to an Interface

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip access-group 110 in
Router(config-if)# exit
```

WARNING

Always end every custom extended ACL with `permit ip any any` (if required) — otherwise the **implicit deny** will silently block ALL other traffic.

7.4 Named ACLs (Alternative, Readable Syntax)

```
Router(config)# ip access-list extended BLOCK_HTTP
Router(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 host 192.168.20.100
    eq 80
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit

Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip access-group BLOCK_HTTP in
```

7.5 Verifying ACLs

```
Router# show access-lists
Router# show ip interface gigabitEthernet0/0
Router# show running-config
```

CHEAT SHEET: ACL Quick Reference

```
STANDARD ACL
  Range: 1-99, 1300-1999
  Filters by: Source IP only
  Place: Close to DESTINATION
  Syntax: access-list <num> {permit|deny} <src-ip> <wildcard>

EXTENDED ACL
  Range: 100-199, 2000-2699
  Filters by: Source IP, Dest IP, Protocol, Port
  Place: Close to SOURCE
  Syntax: access-list <num> {permit|deny} <proto> <src> <wc> <dst> <wc>
       [eq port]

APPLY TO INTERFACE
  ip access-group <num/name> {in|out}

COMMON WILDCARD MASKS
  /24 -> 0.0.0.255      /30 -> 0.0.0.3
  /25 -> 0.0.0.127      host -> 0.0.0.0
  /26 -> 0.0.0.63       any  -> 255.255.255.255 (addr 0.0.0.0)

VERIFY
  show access-lists
  show ip interface <intf>
```

Part III

DHCP & Switch Security

Chapter 8

DHCP Configuration

8.1 What Is DHCP?

The **Dynamic Host Configuration Protocol (DHCP)** automatically gives each host an IP address, subnet mask, default gateway, and DNS server, so that no device has to be configured by hand. DHCP runs over **UDP**: the server listens on **port 67** and the client uses **port 68**.

- Addresses are handed out as **leases** for a limited time.
- A DHCP server can be a dedicated server, or a **router** that runs a DHCP pool.
- Client broadcasts do not cross routers, so a separate **DHCP relay** is needed for other subnets (covered later in this chapter).

8.2 The DORA Process

Every DHCP address assignment follows four messages, remembered as **DORA**.

Step	Message	Purpose
1	DHCPDISCOVER (broadcast)	Client asks for an address
2	DHCPOFFER (unicast/broadcast)	Server proposes an address
3	DHCPREQUEST (broadcast)	Client accepts an offer
4	DHCPACK	Server confirms the lease

NOTE

In Packet Tracer's Simulation Mode, filter the event list to `DHCP` to watch the four messages move between the client and the server.

8.3 Scope Parameters

A DHCP **pool** (scope) defines what the server hands out. The most common parameters are:

Parameter	Command	Example
Excluded addresses	`ip dhcp excluded-address`	192.168.1.1 192.168.1.10
Pool name	`ip dhcp pool NAME`	LAN1
Address range	`network`	192.168.1.0 255.255.255.0
Default gateway	`default-router`	192.168.1.1
DNS server	`dns-server`	8.8.8.8
Lease duration	`lease`	1 (days)

WARNING

Excluded addresses must be configured **before** the pool is created. Put the router's own interface IP and any static servers in the excluded range, or the router may hand them to clients and cause an IP conflict.

8.4 Configuring a Router as DHCP Server

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Topology: Reuse the two-LAN router topology from the LAN chapter.

- LAN1: PC0, PC1 → Switch0 → Router G0/0/0 (192.168.1.1)
- LAN2: PC2, PC3 → Switch1 → Router G0/0/1 (192.168.2.1)

Make sure both router interfaces are configured and active (`no shutdown`) before starting.

CHEAT SHEET: DHCP Server on a Router

```
Router> enable
Router# configure terminal

! 1. Reserve addresses that must never be leased
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10

! 2. Pool for LAN1
Router(config)# ip dhcp pool LAN1
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# lease 1
Router(dhcp-config)# exit

! 3. Pool for LAN2
Router(config)# ip dhcp pool LAN2
Router(dhcp-config)# network 192.168.2.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.2.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# lease 1
Router(dhcp-config)# exit

Router(config)# end
Router# write memory
```

NOTE

`service dhcp` is enabled by default on Cisco IOS, so you normally do not need to type it. If a pool seems to be ignored, check that the `network` statement matches the subnet of the interface the client is connected to.

8.4.1 Setting Up the PCs

1. Click PC0 → **Desktop** → **IP Configuration**.
2. Select the **DHCP** radio button instead of **Static**.

3. Wait a few seconds. The IP address, subnet mask, and default gateway fill in automatically.
4. Repeat for PC1, PC2, and PC3.

```
C:\> ipconfig /all
   ...
   IPv4 Address. . . . . . . . . . . : 192.168.1.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DNS Servers . . . . . . . . . . . : 8.8.8.8
```

TIP

To force a PC to request a fresh lease after a change, run `ipconfig /release` followed by `ipconfig /renew` in its Command Prompt.

8.5 DHCP Relay (ip helper-address)

Broadcast requests from LAN2 cannot reach a DHCP server in LAN1, because the router does not forward broadcasts by default. A **DHCP relay agent** solves this. The router receives the broadcast on the client-facing interface and forwards it as a unicast to the server's address.

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Topology: Add a **Server-PT** to Switch0 on LAN1 (for example, port Fa0/3). Give it the IP address **192.168.1.20** with default gateway **192.168.1.1**.
On the server (Services → DHCP):

1. Turn the DHCP service **On**.
2. Add a pool with the name `LAN2`, default gateway `192.168.2.1`, DNS server `8.8.8.8`, start IP `192.168.2.11`, subnet mask `255.255.255.0`, and a suitable maximum number of users.
3. Click **Add** to save the pool.

CHEAT SHEET: DHCP Relay on the Router

```
Router# configure terminal

! Remove the local LAN2 pool so the router does not answer itself
Router(config)# no ip dhcp pool LAN2

! Forward DHCP broadcasts received on LAN2 to the server
Router(config)# interface gigabitEthernet0/0/1
Router(config-if)# ip helper-address 192.168.1.20
Router(config-if)# end
Router# write memory
```

WARNING

The `ip helper-address` command must be placed on the **client-facing** interface (here G0/0/1), not on the interface closest to the server. Also, the server's pool must match the client's subnet, otherwise it will not offer an address.

8.6 Verification and Troubleshooting

CHEAT SHEET: DHCP Verification Commands

```
Router# show ip dhcp binding          ! which IP went to which MAC, and
    lease time
Router# show ip dhcp pool             ! pool settings and usage counts
Router# show ip dhcp server statistics
Router# show ip dhcp conflict         ! addresses flagged as duplicates
Router# show ip helper-address        ! relay targets (IOS 15+)
```

1. **PC shows 169.254.x.x** → no DHCP offer was received. Check that the pool `network` matches the interface subnet, the interface is `no shutdown`, and the PC is set to **DHCP**.
2. **Client has the wrong default gateway or DNS** → check the pool's `default-router` and `dns-server` lines, and look for a rogue DHCP server (see the next chapter).
3. **Relay not working** → confirm `ip helper-address` is on the client-facing interface, the server is reachable (ping it from the router), and the server has a pool for the client's subnet.
4. **Pool exhausted** → widen the `network` range or shorten the `lease`, then review `show ip dhcp binding`.

Chapter 9

DHCP Snooping

9.1 Why DHCP Snooping?

DHCP has no built-in authentication. Any device on the LAN can answer a DHCP request, which creates two common attacks:

- **Rogue DHCP server:** an attacker runs a DHCP server that hands out a fake default gateway or DNS server, so traffic is redirected through the attacker.
- **DHCP starvation:** an attacker floods the server with requests using fake MAC addresses until the pool is empty, and legitimate users cannot get an address.

DHCP snooping is a Layer 2 switch feature that blocks DHCP replies from untrusted ports and builds a table of which IP address was given to which MAC address on which port.

9.2 Trusted vs Untrusted Ports

Every switch port is **untrusted** by default once DHCP snooping is enabled. Only ports that lead toward the legitimate DHCP server are made **trusted**.

Port Type	Connected To	Allowed to send DHCP server replies?
Trusted	Uplink toward the DHCP server or relay	Yes
Untrusted	End devices (PCs, laptops, unknown hosts)	No

9.3 The DHCP Snooping Binding Table

As the switch watches the DORA exchange on trusted ports, it records each successful lease:

Field	Meaning
MAC Address	Hardware address of the client
IP Address	Address the server leased to that client
Lease (sec)	Remaining lease time
Type	How the entry was learned (dhcp-snooping)
VLAN	VLAN the client belongs to
Interface	Switch port where the client is connected

NOTE

This binding table is also the data source for other security features such as Dynamic ARP Inspection and IP Source Guard. Those features are outside this chapter, but they rely on the same table.

9.4 Configuring DHCP Snooping on a Switch

PACKET TRACER LAB

Topology (Switch-based security test):

- Switch0 (2960) with PC0 and PC1 on Fa0/1 and Fa0/2 (clients).
- A rogue **Server-PT** on Fa0/3 with DHCP service **On**. Give its pool a fake gateway such as 192.168.1.250.
- Switch0 Gi0/1 → Router G0/0/0 (the real DHCP server from the previous section, 192.168.1.1).

Set PC0 and PC1 to **DHCP**. In Simulation Mode, observe that the client may accept an offer from the rogue server. The result depends on which reply arrives first, so you may need to release and renew several times.

CHEAT SHEET: DHCP Snooping on the Switch

```
Switch> enable
Switch# configure terminal

! 1. Turn on DHCP snooping globally
Switch(config)# ip dhcp snooping

! 2. Enable it for the client VLAN (optional, default is all VLANs)
Switch(config)# ip dhcp snooping vlan 1

! 3. Common in lab topologies where no relay agent is used
Switch(config)# no ip dhcp snooping information option

! 4. Trust the uplink toward the real DHCP server
Switch(config)# interface gigabitEthernet0/1
Switch(config-if)# ip dhcp snooping trust
Switch(config-if)# exit

! 5. Rate-limit DHCP requests on client ports (anti-starvation)
Switch(config)# interface range fastEthernet0/1 - 3
Switch(config-if-range)# ip dhcp snooping limit rate 5
Switch(config-if-range)# exit

Switch(config)# end
Switch# write memory
```

WARNING

Forgetting to make the uplink `trusted` is the most common mistake. The switch then drops the real DHCP server's replies too, and **every** client loses its IP address. Always trust the port that leads to the legitimate DHCP server or relay agent.

9.5 Verifying DHCP Snooping

CHEAT SHEET: DHCP Snooping Verification

```
Switch# show ip dhcp snooping              ! status, VLANs, and trusted ports
Switch# show ip dhcp snooping binding      ! MAC-to-IP table built from DHCP
Switch# show ip dhcp snooping statistics   ! packets accepted and dropped
Switch# show running-config interface gigabitEthernet0/1
```

```
Switch# show ip dhcp snooping binding
MacAddress          IpAddress       Lease(sec)  Type            VLAN  Interface
------------------  --------------  ----------  --------------  ----  ------------------
00D0.BA12.3401      192.168.1.11    86400       dhcp-snooping   1     FastEthernet0/1
00D0.BA12.3402      192.168.1.12    86400       dhcp-snooping   1     FastEthernet0/2
```

PACKET TRACER LAB

Test the protection:

1. Enable DHCP snooping as shown above, with Gi0/1 trusted.
2. On PC0 and PC1, run `ipconfig /release` then `ipconfig /renew` several times.
3. Confirm with `ipconfig /all` that every client now uses the **real** default gateway (192.168.1.1) and never the fake one (192.168.1.250).
4. On the switch, run `show ip dhcp snooping binding` to confirm that only the legitimate leases are recorded.

TIP

Use Simulation Mode and filter for `DHCP`. Offers arriving on Fa0/3 (untrusted) should be dropped by the switch, and only offers arriving on Gi0/1 (trusted) should reach the clients.

9.6 Troubleshooting DHCP Snooping

1. **No client gets an address after enabling snooping** → the uplink is not trusted. Run `show ip dhcp snooping` and check the trusted-port list.

2. **Clients still get the fake gateway** → confirm `ip dhcp snooping` is on globally and that the clients' VLAN is listed in `show ip dhcp snooping`.
3. **Rate-limit errors appear** → a legitimate client may be sending too many requests. Increase the `limit rate` value.
4. **Binding table is empty** → clients have not completed DORA through a trusted port, so run a release and renew first.

Chapter 10

Cheat Sheet & Troubleshooting

10.1 Full Command Reference

CHEAT SHEET: Global Commands

```
enable                              ! privileged mode
configure terminal                  ! global config mode
hostname <name>                     ! rename device
exit / end                          ! move up a level
write memory / copy run start       ! save config
show running-config                 ! view active config
show startup-config                 ! view saved config
erase startup-config                ! factory reset (then reload)
reload                              ! restart device
```

CHEAT SHEET: Interface & IP Commands

```
interface <type><number>
ip address <ip> <mask>
no shutdown
description <text>
show ip interface brief
show interfaces
```

CHEAT SHEET: Routing

```
ip route <dest-network> <mask> <next-hop-ip>
show ip route
```

10.2 Common Troubleshooting Checklist

1. **Ping fails** → check IP/mask/gateway on PC via `ipconfig`.
2. **Interface down** → check `no shutdown` was issued; check cable type.
3. **VLAN traffic not passing** → confirm trunk allows the VLAN (`show interfaces trunk`).
4. **Inter-VLAN not working (L3 switch)** → confirm `ip routing` is enabled.
5. **ACL blocking everything** → check for missing `permit any` / implicit deny.

6. **Router-on-a-stick not routing** → verify `encapsulation dot1Q` matches VLAN ID on each sub-interface.

TIP

Use **Simulation Mode** in Packet Tracer and send a PDU between devices to watch exactly where a packet is dropped — this instantly reveals misconfigurations in trunking, routing, or ACLs.